



Final Report

Web Technologies – ICT 1209

**Bachelor of Information and Communication Technology
(BICT)**

Degree Programme

Department of Information and Communication Technology

Faculty of Technology

Rajarata University of Sri Lanka

Mihintale

Details of the Project

Project Title : Habit Tracker

Github repository link : <https://github.com/itt2024062-wq/Habit-tracker->

Submission Date : 2026/10/01

Details of the Group Members

Name	Registration ID	Index No.
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1. Introduction

The Habit Tracker System is a web-based personal productivity application designed to help users create habits, organize them as good or bad habits, monitor daily completion, and view progress over time. The supplied implementation uses PHP for server-side processing, MySQL for persistent storage, Bootstrap for responsive user-interface components, and Chart.js for progress visualization.

The system follows a simple user flow: a user registers an account, signs in, creates habits, views and filters saved habits, records completions, and checks weekly progress and current streak information.

2. Project Overview

2.1 Purpose

The Habit Tracker System is a web-based personal productivity application designed to help users create, manage, and monitor their personal habits. Many people struggle to maintain good habits or break bad ones because they lack a structured way to track their daily progress. Manual methods such as notebooks or spreadsheets are inconvenient and do not provide visual feedback.

This system solves that problem by providing a centralized platform where users can register an account, add habits, classify them as good or bad, record daily completions, and view their progress through charts and statistics. The main purpose is to encourage consistency and self-improvement by making habit tracking simple, visual, and accessible from any browser.

2.2 Target Audience

The primary target audience includes:

- Students who want to build study routines and track daily goals.
- Employees who want to improve productivity and reduce bad habits.
- General users who want to monitor personal routines such as exercise, reading, or avoiding junk food.

The interface is intentionally kept simple so that users with basic computer skills can operate the application without any training.

2.3 Key Features

Feature	Description
User Registration & Login	Creates accounts and authenticates users using PHP sessions and password hashing.
Habit Management	Adds habits with name, type, category and frequency; lists and deletes user-owned habits.
Good/Bad Habit Classification	Separates habits into positive habits and habits the user wants to avoid.
Search & Filtering	Provides client-side search and type filtering on the My Habits page.
Progress Tracking	Stores daily completion records and calculates weekly completion rate and streak.

Progress Visualization	Uses Chart.js to display recent completion activity in a line chart.
Responsive Interface	Uses Bootstrap 5 grid, cards, forms and navigation components for different screen sizes.

3. System Design

3.1 System Architecture

The application follows a lightweight three-layer architecture:

- **Presentation Layer:** The browser renders HTML5, CSS3, Bootstrap 5, and JavaScript. Chart.js is used for visualizations.
- **Application Layer:** PHP scripts handle request processing, input validation, session management, and database queries.
- **Data Layer:** MySQL stores all persistent data including users, habits, and progress records.

This separation keeps the code organized and makes the system easy to maintain and extend.

3.2 Database Schema

The database is named `habit_tracker` and contains three main tables: `users`, `habits`, and `habit_progress`. The relationships are designed using foreign keys with `ON DELETE CASCADE` so that dependent records are automatically removed when a parent record is deleted.

Table 1: users

Field	Type	Description
id	INT (PK)	Primary key, auto-increment
name	VARCHAR(100)	User's full name
email	VARCHAR(150) UNIQUE	Login email address
password	VARCHAR(255)	Hashed password
created_at	TIMESTAMP	Account creation time

Table 2: habits

Field	Type	Description
id	INT (PK)	Primary key, auto-increment
user_id	INT (FK)	References users.id
habit_name	VARCHAR(150)	Name of the habit
description	TEXT	Optional description
category	VARCHAR(100)	Habit category

frequency	VARCHAR(50)	Daily, Weekly, etc.
type	ENUM('Good','Bad')	Habit classification
created_at	TIMESTAMP	Record creation time

Table 3: habit_progress

Field	Type	Description
id	INT (PK)	Primary key, auto-increment
habit_id	INT (FK)	References habits.id
user_id	INT (FK)	References users.id
completed_date	DATE	Date of completion
status	TINYINT	Completion status
created_at	TIMESTAMP	Record creation time

Figure 1: Database relationship diagram derived from database.sql.

3.3 Folder Structure

The project folder is organized as follows:

text

habit_tracker/

```

|
|— config/
|   └─ db.php           (Database connection and session configuration)
|
|— database/
|   └─ database.sql      (Database schema)
|
|— assets/
|   └─ css/
|       └─ style.css     (Shared styling)
|       └─ js/
|           └─ script.js  (Client-side search and filter logic)
|
|— includes/
|   └─ header.php        (Common header)

```

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|   ├── sidebar.php      (Navigation sidebar)
|   └── footer.php       (Common footer)
|
|── public/
|   ├── index.php        (Landing page)
|   ├── about.php        (About page)
|   └── contact.php       (Contact page)
|
|── auth/
|   ├── register.php     (User registration)
|   ├── login.php        (User login)
|   └── logout.php       (Session logout)
|
|── user/
|   ├── dashboard.php    (Logged-in dashboard)
|   ├── add-habit.php    (Create a habit)
|   ├── my-habits.php    (List, search, delete habits)
|   └── progress.php     (Completion and progress tracking)
|
└── README.md

```

4. Implementation

4.1 User Registration and Authentication

Registration (register.php)

The registration page collects the user's name, email, password, and terms acceptance. Server-side validation checks that all fields are filled, the email format is valid, and the password meets the minimum length. The system then checks whether the email already exists in the users table. If the email is new, the password is hashed using `password_hash()` and the record is inserted using a prepared statement.

Login (login.php)

The login page accepts email and password. The system fetches the user by email, verifies the password using `password_verify()`, and stores the user's ID and name in the PHP session. The user is then redirected to `dashboard.php`. Unauthorized users are prevented from accessing authenticated pages by checking the session at the top of each protected script.

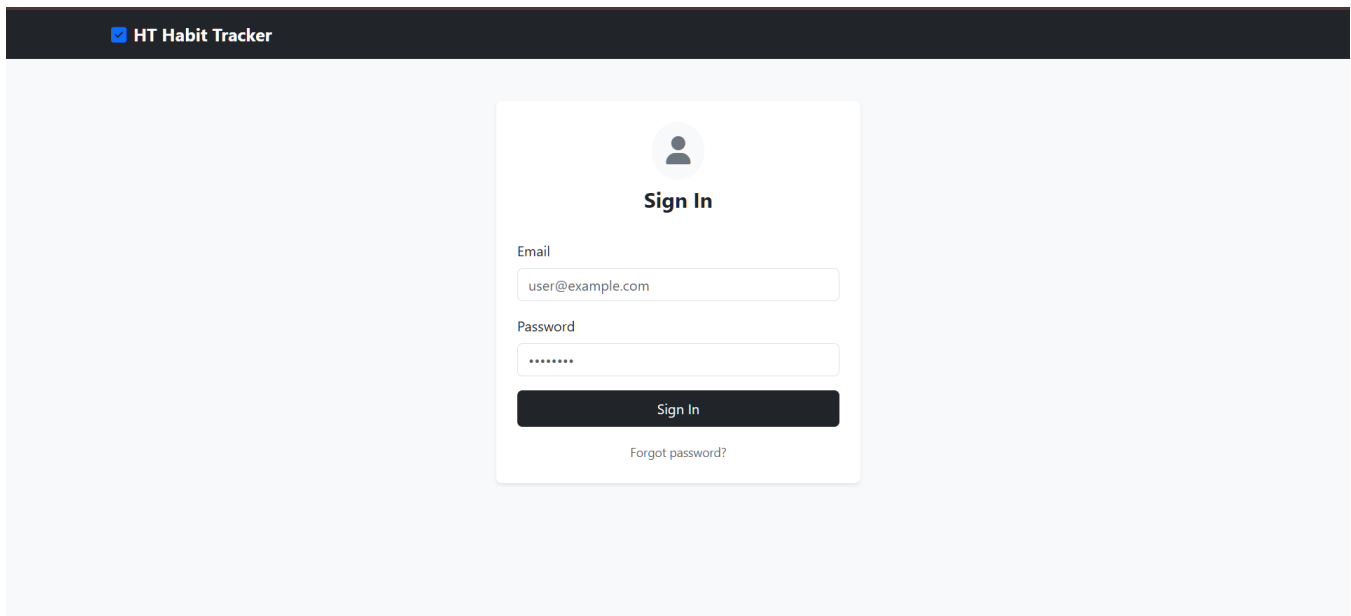


Figure 2: Login interface of the Habit Tracker System.

4.2 Dashboard

The dashboard is the main landing area after login. It uses a fixed left navigation sidebar with links to Home, My Habits, Add Habit, Progress, and Logout. The main content area shows summary cards for total habits, good habits, bad habits, and recent activity. The layout is fully responsive using Bootstrap 5 grid classes.

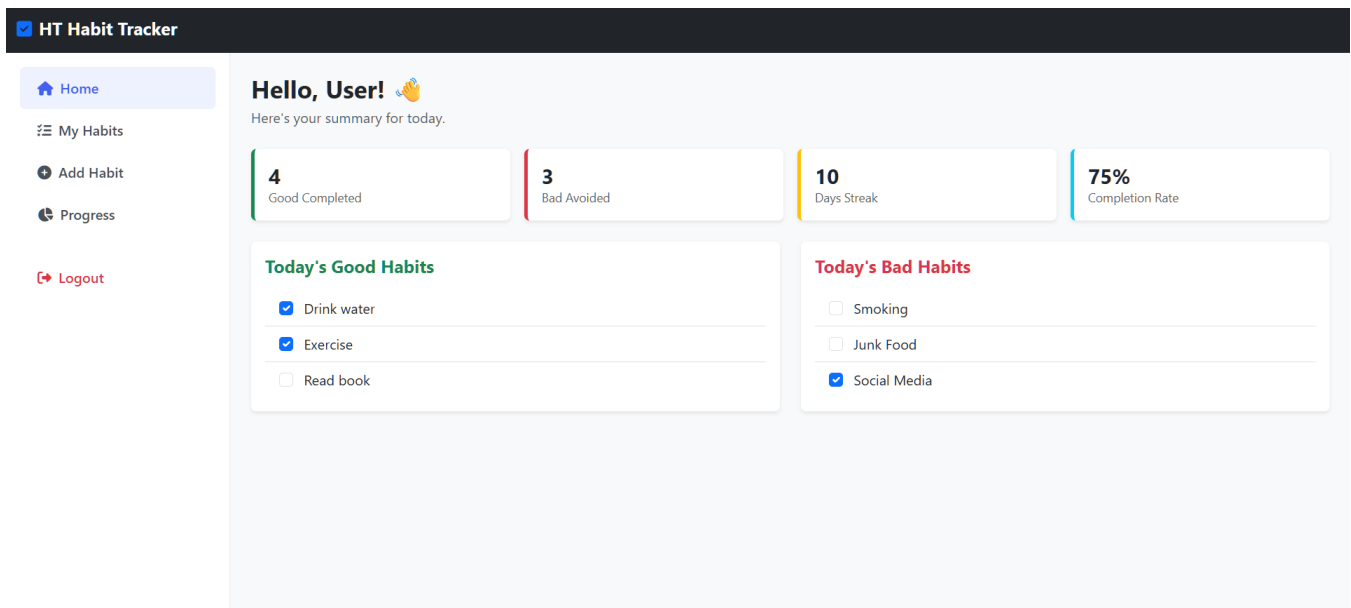


Figure 3: Dashboard interface showing summary cards and navigation.

4.3 Add Habit

The Add Habit module (add-habit.php) allows the user to enter a habit name, select the type (Good or Bad), choose a category, and set the frequency. Server-side validation ensures the habit name and category are provided. A prepared statement inserts the record into the habits table with the current session user ID.

HT Habit Tracker

Home

My Habits

Add Habit

Progress

Logout

Add New Habit

Habit Name:

Enter Habit Name

Habit Type:

☒ Good Habit
 ☐ Bad Habit

Category:

Select Category

Frequency:

Daily

Save Habit

Figure 4: Add New Habit interface with form fields and Save Habit button.

4.4 My Habits

The My Habits page (my-habits.php) retrieves all habits belonging to the current user. Users can search by habit name using a client-side search box, filter between Good and Bad habits using a dropdown, and delete any habit. The delete operation uses both the habit ID and session user ID in the WHERE clause to prevent users from deleting records belonging to others.

HT Habit Tracker

Home

My Habits

Add Habit

Progress

Logout

Manage Habits

Search Habits...

All Types

+ Add Habit

Habit	Type	Action
Drink water	Good	
Exercise	Good	
Read book	Good	
Smoking	Bad	
Junk Food	Bad	
Social Media	Bad	

Figure 5: Manage Habits page showing search, filter, and delete options.

4.5 Progress Tracking

The Progress page (progress.php) is the most interactive part of the system. When the user marks a habit as completed, the script:

1. Checks whether a completion record already exists for the same habit, user, and date. If yes, it prevents duplicates.

2. Inserts a new row into habit_progress.
3. Calculates the weekly completion rate based on the last seven days.
4. Calculates the current consecutive-day streak.
5. Passes the data to Chart.js to render a line chart showing weekly performance.

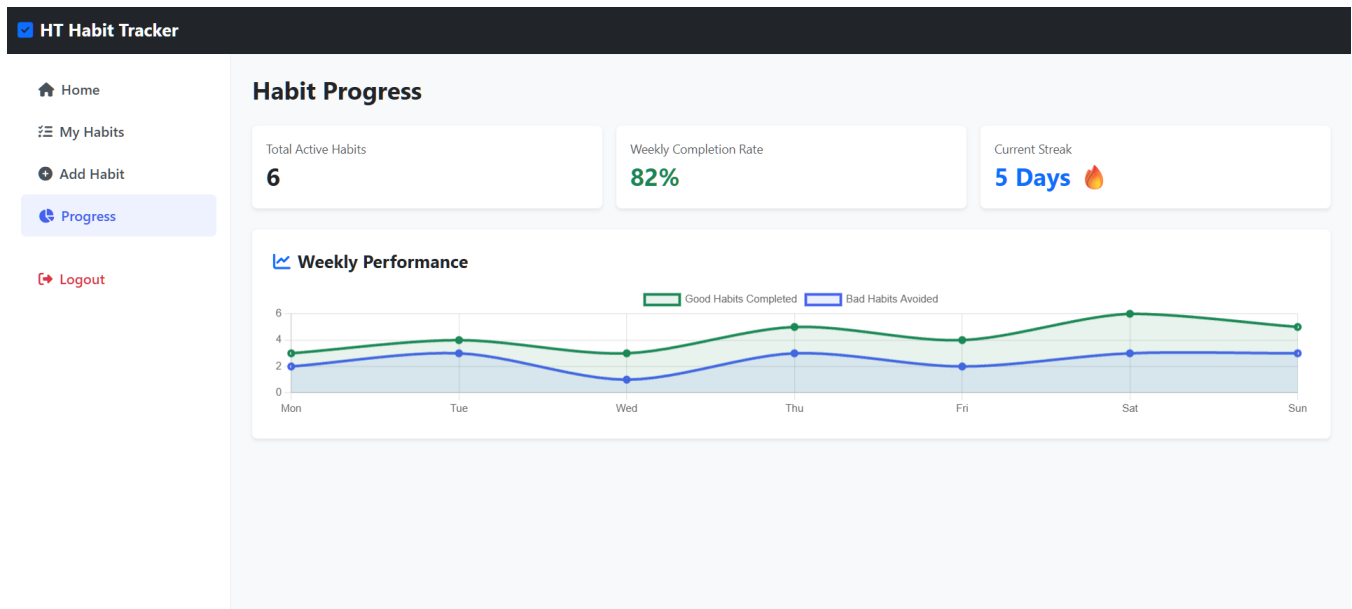


Figure 6: Habit Progress page showing weekly completion rate, current streak, and Chart.js line chart.

5. Challenges and Solutions

Challenge	Problem	Solution
Maintaining user-specific data	Queries could accidentally expose records belonging to another user.	Every SELECT, INSERT, and DELETE query includes the logged-in user_id from the session.
Secure password handling	Plain-text password storage would create a serious security risk.	Passwords are stored using password_hash() and verified using password_verify().
Duplicate daily completion	Repeated clicks could create multiple records for one habit on the same day.	progress.php checks for an existing record for the same habit, user, and date before inserting a new completion.
Database consistency	Deleting a habit could leave orphaned progress rows.	Foreign keys use ON DELETE CASCADE so dependent progress records are removed automatically.
Usable navigation	Users need consistent access to major functions after login.	All authenticated pages share a common left-side navigation sidebar.
Progress calculation	Weekly completion and streak values must reflect stored data accurately.	The PHP implementation dynamically calculates the last seven dates, counts completions, and tracks consecutive completion days.

Responsive design	The application must work on both desktop and mobile screens.	Bootstrap 5 grid and components are used throughout the interface.
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6. Individual Contribution

The following table shows the contribution breakdown for this project:

Member	Main Responsibility	Contribution
ITT/2024/029 & ITT/2024/062	Backend & Database	PHP authentication, database connection, CRUD operations, progress calculations, and database.sql.
	Frontend & UI	HTML pages, Bootstrap layout, navigation/sidebar, forms, and responsive styling.
	Testing & Documentation	Functional testing, UI testing, screenshots, report preparation, and presentation support.

7. Setup Instructions – XAMPP

Follow these steps to run the Habit Tracker System on a local machine using XAMPP:

Step 1: Install XAMPP with Apache, MySQL, and PHP.

Step 2: Extract the project folder so that the application is placed inside:

text

C:\xampp\htdocs\habit_tracker

Step 3: Start Apache and MySQL from the XAMPP Control Panel.

Step 4: Open phpMyAdmin in a browser:

text

http://localhost/phpmyadmin

Step 5: Create/import the database by selecting the supplied database.sql file and executing it. The script creates the habit_tracker database and its users, habits, and habit_progress tables.

Step 6: Check config/db.php and make sure the MySQL host, username, password, and database name match the local XAMPP configuration.

Step 7: Open the application in the browser:

text

http://localhost/habit_tracker/public/index.php

Step 8: Register a new user, sign in, add habits, and use the Progress page to test daily completion and statistics.

7.1 Database Import Summary

The database.sql file uses MySQL/InnoDB and utf8mb4. It creates the habit_tracker database if it does not already exist, then creates the three application tables and their indexes/foreign keys.

7.2 Recommended Test Sequence

1. Register a test account.
 2. Log in with the new account.
 3. Create at least two habits (one Good and one Bad).
 4. Open My Habits and test search/filter/delete.
 5. Open Progress and mark a habit as completed.
 6. Refresh Progress and verify the completion statistics and chart.
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8. References

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